



Analytical Methods

Determination of aloe emodin in *Aloe vera* extracts and commercial formulations by HPLC with tandem UV absorption and fluorescence detectionRoberto Mandrioli^{a,*}, Laura Mercolini^a, Anna Ferranti^a, Salvatore Fanali^b, Maria Augusta Raggi^a^a Laboratory of Pharmaco-Toxicological Analysis, Department of Pharmaceutical Sciences, Alma Mater Studiorum – University of Bologna, Via Belmeloro 6, I-40126 Bologna, Italy^b Institute of Chemical Methodologies, Italian National Council of Research, Area della Ricerca di Roma 1, I-00015 Monterotondo Scalo, Rome, Italy

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ABSTRACT

Aloe emodin is an anthraquinone compound found in *Aloe vera* and other species of the Asphodelaceae and the Polygonaceae families, which has recently attracted much attention as a prospective antineoplastic agent. A HPLC method with tandem UV absorption and fluorimetric detection, was developed and validated for the analysis of aloe emodin in products obtained from *Aloe* leaves, such as capsules, tablets, dried extracts and mother tinctures. The stationary phase was a C18 reversed-phase column and the mobile phase was composed of water and methanol (30/70, v/v). Satisfactory linearity was obtained over the 10.0–1000.0 ng/mL range for photodiode array detection (limit of detection: 3 ng/mL) and over the 2.5–1000.0 ng/mL range for fluorimetric detection (limit of detection: 0.8 ng/mL). Aloe emodin levels were determined in *A. vera* extracts and commercial formulations by both detection means, with good precision (R.S.D. < 9%) and accuracy (recovery > 85%) and consistent results. Thus, the method seems to be suitable for the analysis of aloe emodin in different herbal and commercial products; it could also be useful for the identity confirmation of formulations and extracts.

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1. Introduction

Aloe is a genus including about four hundred species of succulent plants, probably originating from Africa. The most important and well-known species of this genus is *Aloe vera* (also known as *Aloe barbadensis* Mill.), a stemless or very short-stemmed plant growing up to about one metre tall. The leaves are green, with some varieties showing white flecks. It is now naturalised to a very wide range of regions, including temperate and tropical areas of China, India, Pakistan, Southern Europe, Australia, Central America and the USA. This plant has been used as a medicine for more than three thousand years, being cited in a papyrus from the 16th century BCE, as well as in Dioscorides' *De Materia Medica* and Pliny the Elder's *Natural History* (1st century CE), and is part of traditional medicinal practices of China, Japan, Russia, South Africa, USA, Jamaica and India (Boudreau & Beland, 2006). *A. vera* extracts are widely used both in herbal medicine and in cosmetics and toiletries; they are used externally, as emollients, moisturisers and cicatrisants, and internally, as digestive and laxative agents. Many other beneficial effects have been attributed to *A. vera* extracts and in particular to the exudate seeping out from cut leaves; among them, there are claims of antibacterial, antifungal and antiviral activities (Vogler & Ernst, 1999); reduction of glycaemia (Yongchaiyudha, Rungpitarangsi, Bunyapraphatsara, &

Chokechaijaroenporn, 1996), hyperlipidaemia and psoriasis symptoms (Choonhakarn, Busaracome, Sripanidkulchai, & Sarakarn, 2010).

The *A. vera* exudate contains numerous biologically active compounds, the most well known of which is aloin, a drastic laxative. According to some authors, this effect of aloin would be due to its biotransformation in the gut to aloe emodin (1,8-dihydroxy-3-(hydroxymethyl)-9,10-anthracenedione, AEM, Fig. 1a), which would be the real active principle. AEM is one of the most important anthraquinones found in several species of *Aloe* and in the leaves and roots of other plants. AEM has recently attracted some attention due to its possible anticancer activity (Fenig, Nordenberg, Beery, Sulkes, & Wasserman, 2004); in particular, it was found to inhibit *N*-acetyltransferase activity and gene expression in human malignant melanoma cells (Lin et al., 2005) and to have anti-proliferative and pro-apoptotic effects on gastric (Guo, Xiao, Liu, Gong, & Le, 2008) and tongue (Chiu et al., 2009) cancer cells. If confirmed, these results could lead to trials of AEM (or closely related analogues) as a prospective antineoplastic agent (Pecere et al., 2000). Anyway, the mechanisms of its anticancer activity have not yet been completely explained.

Aim of this study is the development of reliable and feasible analytical methods for the quantification of AEM in crude *Aloe* extracts and in formulations. The determination of AEM can be useful for quality control, in order to avoid adulterations and to certify the activity of the sample. On the other hand, it can also be important

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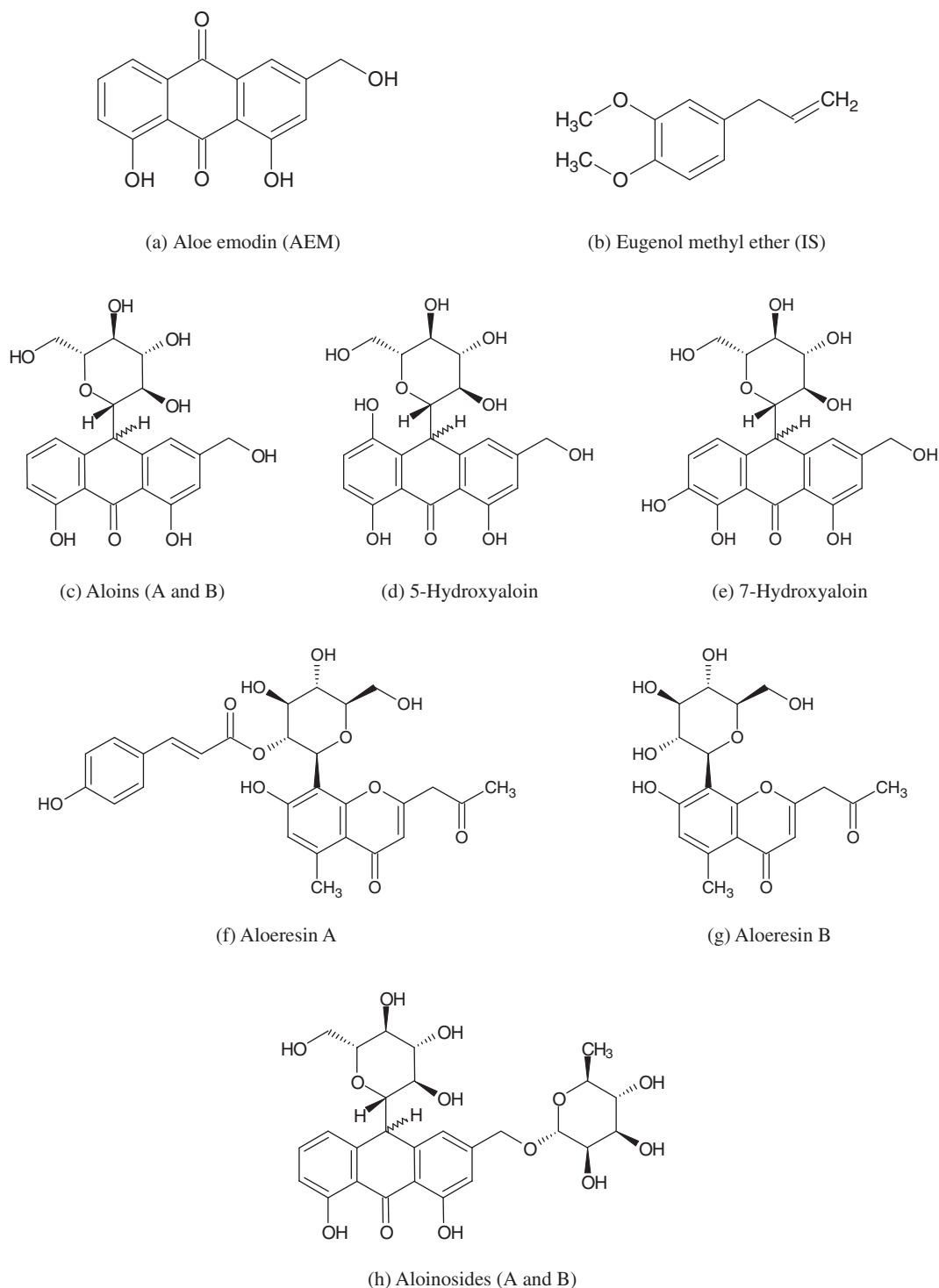


Fig. 1. Chemical structures of (a) aloe emodin (AEM), (b) eugenol methyl ether (IS) and (c–h) the *Aloe* exudate constituents used to test the method selectivity.

for safety purposes, due to the possible cytotoxic activity of AEM; its accurate dosage in formulations and extracts intended for other purposes can avoid important side and toxic effects.

Some analytical methods can be found in the literature for the analysis of AEM in different plants and herbs, such as rhubarb (Tian, Zhang, Chen, & Hu, 2006; Koyama, Morita, & Kobayashi, 2007; Lin, Wu, Lin, & Sheu, 2006; Ye, Han, Chen, Zheng, & Guo, 2007; Tian, Wang, Chen, Chen, & Hu, 2007) and other *Polygonaceae* (Avula, Joshi, Wang, & Khan, 2007; Jiao & Zuo, 2009; Tian et al., 2006; Yi, Zhang, & Cai, 2007; Zuo, Wang, Lin, Guo, & Deng, 2008)

and senna (Panichayupakaranant, Sakunpak, & Sakunphueak, 2009; Fernand et al., 2008; Jiang, Lv, & Wang, 2005). To identify and quantify AEM in *Aloe* extracts, GC–MS (Loots, Van der Westhuizen, & Botes, 2007), HPLC–UV (Okamura, Asai, Hine, & Yagi, 1996; Park et al., 1998) and capillary electrophoresis (Yang, Wang, Guo, & Chen, 2004) have been used. The methods, which have been applied to *Aloe*, generally analyse many compounds of the same family (e.g., phenols) using complicated and expensive procedures such as gradient elution (Okamura et al., 1996; Park et al., 1998) or expensive instrumentations such as GC–MS (Loots

et al., 2007). On the other hand, capillary electrophoresis (Yang et al., 2004) has low sensitivity, which can prevent the proper quantification of the analyte when present at low concentrations, or require concentration prior to analysis. Finally, none of these methods have been applied to the quality control of formulations containing *Aloe* extracts. Aim of this study was the development of a fast and feasible analytical method, based on isocratic HPLC, for the analysis of AEM in both crude extracts and formulations, with high sensitivity and selectivity.

2. Materials and methods

2.1. Chemicals and solutions

AEM, eugenol methyl ether used as the internal standard (1,2-dimethoxy-4-prop-2-enylbenzene, IS, Fig. 1b) for the control of retention times and HPLC grade methanol were from Sigma–Aldrich (St. Louis, USA). Ethyl acetate for analysis was purchased from Carlo Erba (Milan, Italy). Ultrapure water (18.2 MΩ cm) was obtained by means of a MilliQ apparatus by Millipore (Milford, USA).

Stock solutions of the analyte and the IS (1 mg/mL) were prepared by dissolving suitable amounts of each pure substance in methanol and were stable for at least two months when stored at -20°C (as assessed by HPLC assays). Standard solutions were prepared fresh every day by diluting stock solutions with the mobile phase and directly injected into the HPLC system.

2.2. Instrumentation and chromatographic conditions

The chromatographic system was composed of a Jasco (Tokyo, Japan) PU-1580 chromatographic pump, a Jasco MD-910 photodiode array (PDA) detector and a Varian Mod. 9075 spectrofluorimetric detector; for quantitative purposes, the PDA detector was set at 254 nm, while the spectrofluorimetric detector was set at $\lambda_{\text{exc}} = 435\text{ nm}$, $\lambda_{\text{em}} = 515\text{ nm}$. The distance between the two detector cells was 21 cm (corresponding to a retention volume of 0.10 mL or a retention time of 6 s).

Separations were obtained on a Phenomenex (Torrance, CA, USA) Gemini C18 reversed-phase column ($250 \times 4.6\text{ mm I.D.}$, $5\text{ }\mu\text{m}$) coupled to a Phenomenex C18 guard column ($3 \times 4.6\text{ mm I.D.}$, $5\text{ }\mu\text{m}$) and kept at room temperature. The mobile phase was composed of a methanol/water (70/30, v/v) mixture, degassed by an ultrasonic bath. The flow rate was 1 mL/min and the injections were carried out through a 50- μL loop. Data processing was handled by means of a Varian (Walnut Creek, USA) Star Chromatography 4.0 software. A Hettich (Tuttlingen, Germany) Universal 32 R centrifuge was used.

2.3. Sample preparation and pre-treatment

2.3.1. Preparation of “lab” extracts

One basal leaf of an *A. vera* plant was cut and suspended, with the cut end facing down, over a previously weighed beaker. After 24 h, the exudate collected at the bottom of the beaker was dried in an oven (at 50°C for 1 h) and the beaker was weighed again. The dried extract was then removed with a spatula, finely ground and transferred into a stoppered glass container.

2.3.2. Pre-treatment of plant extracts

Ten milligrams of dried plant extract (either the “lab” extract obtained by the authors in laboratory or *Aloe* “shop” extracts bought in herbalist’s shops) were accurately weighed and transferred into a polypropylene vial; 500 μL of methanol were added and the mixture was vortex-mixed for 5 min and centrifuged

($4000 \times g$) for 5 min. The supernatant was subjected to a preparative TLC procedure as follows. Fifty microlitres of the clear supernatant were deposited in a narrow horizontal line on a $5 \times 10\text{ cm}$ TLC silica gel 60 F₂₅₄ plate (Merck, Darmstadt, Germany), at 1 cm from the bottom edge, using a micropipette. The plate was vertically developed up to a height of 8 cm in a previously conditioned, closed chamber with a mobile phase composed of petroleum ether/ethyl acetate (60/40, v/v). AEM, evidenced as a dark stain by UV at 245 nm, was scratched out and the silica transferred into a polypropylene vial. Ten microlitres of IS standard solution and 200 μL of methanol were added and thoroughly mixed by vortex. The vial was then centrifuged for 3 min at $4000 \times g$ and the supernatant was injected into the HPLC system (after suitable dilution with the HPLC mobile phase, if necessary).

2.3.3. Pre-treatment of commercial formulations

The following commercial formulations were analysed: two different kinds of tablets, one kind of capsules and one kind of mother tincture. The content of five capsules or tablets was weighed, finely ground in a mortar, thoroughly mixed and transferred into a stoppered glass container. A powder amount equivalent to the mean content of two capsules or tablets was accurately weighed and transferred into a volumetric flask; 2 mL of methanol were added and the mixture was vortex-mixed for 5 min and centrifuged ($4000 \times g$) for 5 min. Fifty microlitres of the clear supernatant were then subjected to the preparative TLC procedure described above (see Section 2.3.2) and injected into the HPLC system after suitable dilution with the HPLC mobile phase, if necessary.

Liquid mother tincture of *A. vera* was diluted 1:50 with methanol and 50 μL of this solution were subjected to the same preparative TLC procedure and HPLC analysis as above.

2.4. Method validation

All method validation assays were carried out according to the United States Pharmacopeia (USP32) guidelines (United States Pharmacopeial Convention 2009).

2.4.1. Calibration curves

Analyte standard solutions (prepared daily) at seven different concentrations containing the IS at a constant concentration (5 $\mu\text{g/mL}$) were injected into the HPLC system. The procedure was carried out in triplicate for each concentration. The analyte peak areas (arbitrary unites) obtained were plotted against the corresponding concentrations of the analyte (expressed as ng/mL) and the calibration curves were set up by means of the least-square method. The values of Limit of Quantification (LOQ) and Limit of Detection (LOD) were calculated as the analyte concentrations, which give rise to peaks whose heights are 10 and 3 times the baseline noise, respectively.

2.4.2. Extraction yield (absolute recovery)

To evaluate extraction yields, the following procedure was used. Each sample was subjected to the extraction procedure described in Section 2.3 and analysed. Then, the same material was subjected a second time to the extraction procedure and analysed. If this second extraction step produced a quantifiable analyte peak, the procedure was repeated again until no quantifiable AEM peak was detected. The extraction yield of each single extraction procedure was calculated by comparing the AEM amount found in each extraction with the sum of all analyte amounts found in all extractions.

2.4.3. Precision

The assays described under “Extraction yield” were repeated six times within the same day to obtain repeatability (intraday preci-

Table 1
Linearity parameters on standard solutions.

Detection	Linearity range (ng/mL)	Linearity equation $y = ax + b^a$		r^2	LOQ (ng/mL)	LOD (ng/mL)
		a	b			
UV absorbance	10.0–1000.0	0.0110	0.0007	0.9999	10.0	3.0
Fluorimetric	2.5–1000.0	0.154	0.004	0.9998	2.5	0.8

^a x is the analyte concentration, expressed as ng/mL; y is the peak area, expressed as arbitrary units.

sion) and six times over six different days to obtain intermediate precision (interday precision), both expressed as R.S.D.% values.

2.4.4. Selectivity

Standard solutions of several different compounds found in *Aloe* species were injected; if the resulting chromatograms contained any interference peak, the potentially interfering compounds were subjected to the extraction procedure and injected to ascertain if they could be extracted.

2.4.5. Accuracy

Accuracy was evaluated by means of recovery assays. Known amounts of analyte standard solutions (corresponding to the lower limit, a middle level and a high level of the calibration curve) were added to samples whose AEM content was already known (i.e., which had been already analysed). After extraction and analysis, the recovery of the added analyte was calculated. The assays were repeated three times during the same day to obtain mean recovery and SD data.

3. Results and discussion

3.1. Choice of the experimental conditions

AEM displays high absorbance at 254 nm and also possesses native fluorescence at 515 nm, when excited at 435 nm. For this reason, both UV absorption and fluorimetric detection could be coupled to HPLC to obtain suitably sensitive and selective assays. So, it was decided to connect both a PDA and a fluorimetric detector, in series, to the HPLC system. This set-up confers a higher degree of selectivity and reliability to the assay; PDA detection can also be used to assess peak purity.

Since AEM does not contain easily ionisable groups (only very weakly acidic phenol groups), water/organic solvent mixtures were tested as possible mobile phases. Both water/methanol and water/acetonitrile mixtures proved to be suitable for analyte elution, however better peak shapes were obtained when using methanol as the organic modifier. The methanol percentage was varied between 50% and 80%; high methanol percentages led to shorter run times, but also to a method more prone to interference due to other endogenous compounds co-eluting with AEM. A methanol percentage of 60% demonstrated to be a good compromise between satisfactory retention and separation from early-eluting endogenous compounds. Eugenol methyl ether was chosen as the internal standard (IS) because, under these conditions, it was eluted in a short time and furthermore, it is a natural compound not found in *Aloe* plants or extracts, thus no interference from endogenous eugenol methyl ether was possible.

Neat, symmetric and well-separated peaks are obtained under the reported conditions, with run times shorter than 9 min.

Since the IS is not fluorescent, it cannot be detected by spectrofluorimetry: in those cases, when a fluorescent IS is needed, trazodone can be used. Trazodone ($t_R = 9.4$ min), however, is more strongly retained than AEM, thus run times become longer when introducing this alternative IS.

3.2. Assay linearity

Satisfactory linearity ($r^2 > 0.9995$) was obtained over the 2.5–1000 ng/mL concentration range for fluorimetric detection and over the 10.0–1000.0 ng/mL range for UV detection. LOQ and LOD values, as well as linearity parameters, are reported in Table 1 for both detection means.

Intraday precision (repeatability) and interday precision (intermediate precision) assays gave good results: R.S.D. values were always lower than 7.0% for fluorimetric detection and lower than 6.2% for UV detection.

3.3. Development of the sample pre-treatment procedure

Regarding *Aloe* extracts, it has been reported (Kiridena, Poole, Miller, & Poole, 1995) that AEM can be separated from most of the other endogenous *Aloe* compounds and identified by TLC on silica plates using petroleum ether/ethyl acetate as the mobile phase. Thus, a relatively fast and feasible preparative TLC procedure was chosen as a pre-treatment step to obtain clean extracts and good extraction yields. Preliminary assays showed that the R_f of AEM increased when increasing the petroleum ether percentage in the TLC mobile phase (in the 30–80% range); 60% was found to be the best value to separate AEM from most similar compounds (such as aloins), while achieving acceptable development times on a 10-cm TLC plate (R_f of AEM = 0.77).

Methanol was found to be very efficient in extracting AEM from silica (after removing it from the plate). This is important, since the extract is directly injected into the HPLC system; so, a large volume of solvent would cause an unnecessary dilution of the analyte. Preliminary assays with single and multiple extractions, using methanol volumes in the 50–1000 μ L range, were carried out. It was found that multiple extractions are not needed, since the overall extraction yield (including the first extraction from the exudate and the re-dissolution from the silica) was always higher than 95%. No significant yield increase was found when increasing the extraction volume beyond 200 μ L; this volume was chosen for all assays in order to avoid unnecessary analyte dilution, as already noted. In fact, different *Aloe* species and formulations have very different AEM contents, which are of course unknown before analysis; for this reason, it is usually more prudent if the solution resulting from the sample pre-treatment is very concentrated, increasing the sensitivity of the assay. If the AEM concentration is very high, the solution can be diluted with the mobile phase.

3.4. Method validation

Extraction yield (absolute recovery) and precision assays were carried out on extracts and on commercial formulations as described in Section 2.4.2.

The results of these assays are reported in Table 2. As one can note, mean extraction yields were always higher than 95%. Precision results were also satisfactory: R.S.D. values were always lower

Table 2

Extraction yield and precision results on real samples.

Detection	Matrix	Mean extraction yield (%) ^a	Precision (RSD%) ^a	
			Repeatability	Intermediate precision
UV absorbance	“Lab” extract	95.5	6.8	8.2
	“Shop” extract	96.3	6.6	8.0
	Tablets 1	96.1	4.3	4.8
	Tablets 2	96.5	4.5	4.7
	Capsules	95.8	5.1	5.9
	Mother tincture	95.3	6.0	7.4
Fluorimetric	“Lab” extract	95.3	7.1	8.9
	“Shop” extract	96.1	7.0	8.3
	Tablets 1	96.1	5.0	6.4
	Tablets 2	96.2	5.0	6.3
	Capsules	95.7	5.7	6.6
	Mother tincture	95.1	6.8	7.8

^a n = 6.

than 7.2% for repeatability and lower than 9.0% for intermediate precision.

Selectivity was evaluated by injecting into the HPLC standard solutions of several *Aloe* exudate constituents, namely: aloin A and B (Fig. 1c), 5-hydroxyaloin A (Fig. 1d), 7-hydroxyaloin A (Fig. 1e), aloeresin A (Fig. 1f) and B (Fig. 1g), aloinoside A and B (Fig. 1h). They were not retained or had very short retention times (less than 4 min), thus none of these compounds could cause any interference in the analysis.

3.5. Analysis of real samples

The validated method was applied to the determination of AEM in plant extracts and in commercial products; some examples of the results obtained from these analyses are shown in Figs. 2 and 3.

Fig. 2a–b shows the chromatograms of a “lab” extract prepared by the authors from an *A. vera* plant, obtained with UV absorption (a) and fluorimetric (b) detection (1:100 sample dilution). The AEM concentrations found in this real sample were identical for the two detection means, equal to 1.58 mg/g of dried extract. No interfering peak is apparent.

In a similar way, the chromatograms shown in Fig. 3a–b were obtained from the “mother tincture” formulation (no declared extract content). In this case, the AEM levels found were 0.45 mg/mL.

PDA detection allows the use of online UV spectra to assess peak purity; in all cases, the peak with a retention time corresponding to that of AEM resulted to be within the peak purity limits. This confirms the peak identification and excludes the presence of significant interference.

Accuracy was evaluated by means of recovery assays. Standard solutions of the analytes at three different concentrations (low, middle and high level) and of the IS at a constant concentration were added to samples containing known amounts of AEM (i.e., samples, which had already been analysed). Then, the recovery of the added analyte was calculated, as well as the standard deviation of the assays. The results of the accuracy assays are reported in Table 3: mean recovery values were always higher than 85%. Thus, method accuracy is satisfactory.

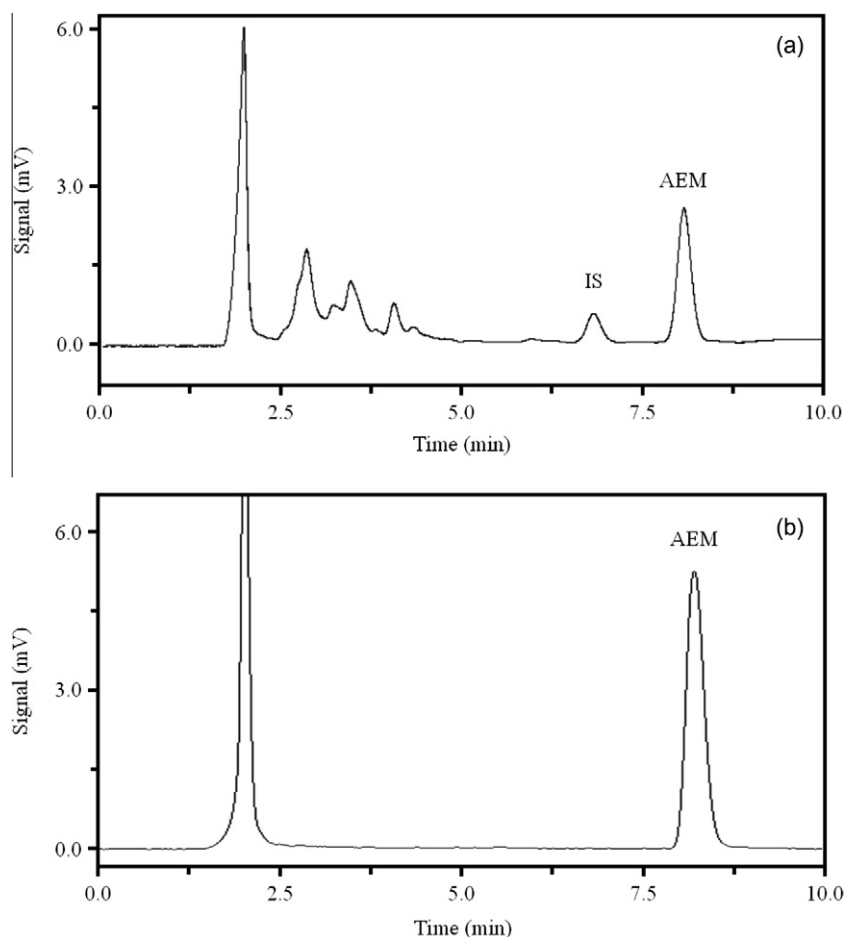


Fig. 2. Chromatograms of an *Aloe vera* “lab” extract (prepared by the authors): (a) UV absorption and (b) fluorimetric detection. Chromatographic conditions: stationary phase, Gemini C8 reversed-phase column (250 × 4.6 mm I.D., 5 μm); mobile phase, methanol/water (70/30, v/v); flow rate, 1.0 mL/min; injection volume, 50 μL.

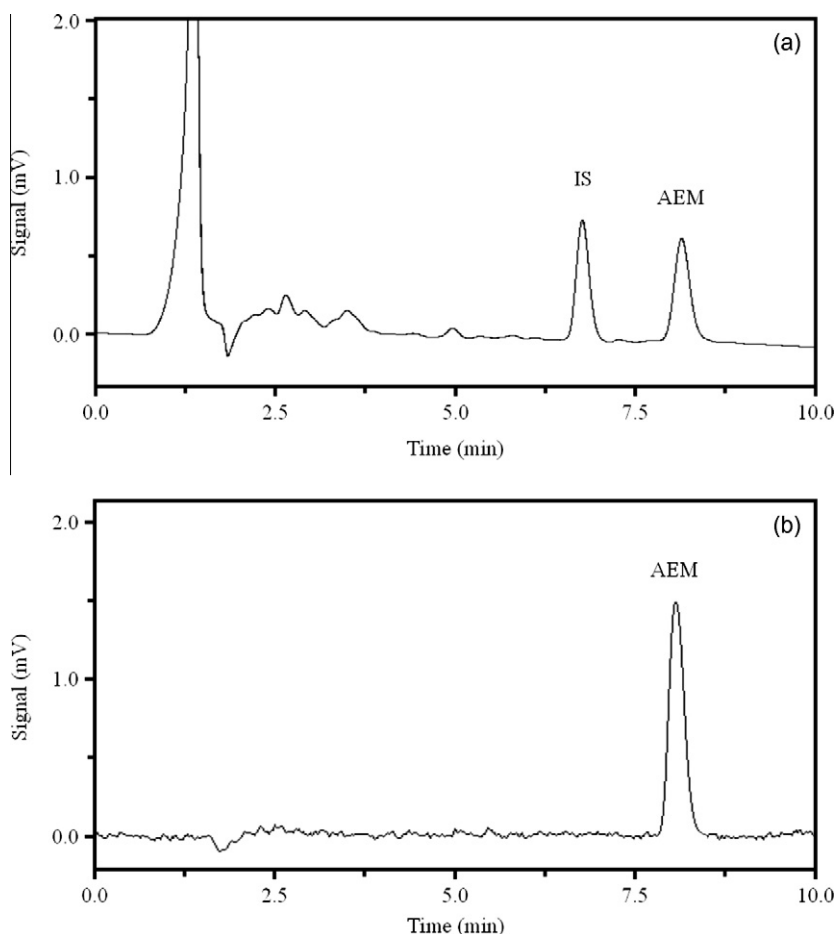


Fig. 3. Chromatograms of a mother tincture formulation declaredly obtained from *Aloe vera*: (a) UV absorption and (b) fluorimetric detection. Chromatographic conditions: as in Fig. 2.

Table 3
Accuracy assays on real samples.

Detection	Concentration (ng/mL)	Mean recovery (%) ^a	SD ^a
UV absorbance	10	85.3	7.4
	250	87.2	5.1
	500	86.3	2.1
Fluorimetric	2.5	85.5	7.9
	250.0	87.1	5.5
	500.0	85.9	4.4

^a $n = 3$.

4. Conclusion

An analytical method, based on HPLC with both UV absorption and fluorimetric detection in series, has been developed and validated for the determination of the AEM content of *Aloe* extracts and derived formulations. The method is feasible and shows good sensitivity and precision. It includes a sample pre-treatment procedure, based on solvent extraction and preparative TLC, which grants very good selectivity to the assay. The coupling of UV absorption and fluorimetric detection allows the comparison of the results obtained with the two detection means and this grants a higher level of confidence to the results.

With respect to previously published methods, the proposed one has the advantage of having been validated also for the analysis of AEM content in formulations; moreover, it uses relatively inexpensive instrumentations (HPLC–UV and HPLC–F rather than

GC–MS, Loots et al. 2007) and has better sensitivity (up to 1000 times) when compared to capillary electrophoresis (Yang et al., 2004). Finally, it does not use gradient elution or multiple solvent extractions (Okamura et al., 1996; Park et al., 1998), with obvious advantages in terms of costs, feasibility and environmental compatibility.

In conclusion, the developed method seems to be suitable for the control of the AEM content of *Aloe* extracts and commercial formulations.

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